

Retrieval-Augmented Clinical Decision Support for Antibiotic Prescribing: Integrating LLM Reasoning with AIFA Guidelines and Pharmaceutical Data

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1 Introduction

Antibiotic prescribing in primary care requires clinicians to integrate patient-specific information with clinical guidelines and local epidemiological considerations. This decision is particularly relevant in the context of antimicrobial resistance (AMR), for which inappropriate antibiotic use represents an important contributing factor. General practitioners (GPs) must therefore make prescribing decisions efficiently while considering heterogeneous sources of information, including clinical presentation, patient characteristics, and current evidence.

In recent years, large language models (LLMs) have shown considerable potential for processing and reasoning over complex medical information. However, the use of LLMs in clinical contexts also presents important challenges, including the reliability of generated recommendations, the potential for hallucinated information, and the difficulty of ensuring that generated outputs are grounded in authoritative medical sources.

Additionally, existing research on AI for clinical decision support has predominantly focused on hospital-based environments and on supporting healthcare professionals working with relatively complex clinical workflows. In contrast, GPs operate in a different setting, where consultations are often shorter and decisions must be made using the information available at the point of care.

This project addresses this gap by investigating the development of a RAG-based clinical decision-support system, based on official data from the Italian Medicines Agency (AIFA), for antibiotic prescribing in primary care.

The system is subsequently examined through a qualitative assessment involving medical experts, with the aim of understanding its potential usefulness and identifying limitations rather than demonstrating autonomous clinical decision-making.

2 Research Question and Methodology

This project investigates whether a RAG system can be used to provide evidence-grounded antibiotic prescribing support in primary care. The research question is:

Can a RAG-based clinical decision-support system combine patient-specific information, retrieved clinical evidence, regional antibiotic pressure, and pharmaceutical data to generate clinically relevant and traceable antibiotic recommendations for general practitioners?

To address this question, the project focuses on the development of an end-to-end prototype rather than on benchmarking model performance. The system is designed to (i) retrieve relevant clinical evidence for a given patient case, (ii) incorporate patient and regional information into the LLM context, (iii) generate a structured antibiotic recommendation with supporting reasoning, and (iv) link the identified active ingredient to corresponding pharmaceutical products in AIFA data.

2.1 Vector Database Creation

The knowledge base used by the RAG system was constructed from the *Manuale antibiotici AWaRe* (*Access, Watch, Reserve*), the Italian edition of the WHO AWaRe Antibiotic Book prepared and published by the Italian Medicines Agency (AIFA) https://www.aifa.gov.it/documents/20142/1811463/Manuale_antibiotici_AWaRe.pdf. The document provides evidence-based recommendations for the management of common infectious syndromes, including indications for antibiotic treatment and guidance on antibiotic selection in both primary and hospital care.

Only the portion of the document covering pages 35–240 was retained, corresponding to the sections relevant to the primary-care clinical scenarios considered in this project. These include, among others, bronchitis, acute otitis media, pharyngitis, acute sinusitis, bacterial eye infections, and lower urinary tract infections.

The extracted text was divided into overlapping chunks using a recursive character-based splitting strategy, with a chunk size of 900 characters and an overlap of 150 characters. The overlap was introduced to reduce the risk of separating clinically relevant information across consecutive chunks and to preserve contextual continuity during retrieval.

Each chunk was subsequently converted into a vector representation using a multilingual sentence embedding model. Two candidate models were initially considered: `paraphrase-multilingual-mpnet-base-v2` and `multilingual-e5-base`. A preliminary retrieval test was performed using a mockup clinical query: the comparison showed that MPNet produced less reliable semantic matches, including confusion between clinically distinct terms such as *congintivite* and *farin-gite*. The multilingual E5 model instead consistently retrieved passages relevant to the query and was therefore selected for the final RAG pipeline.

The selected E5 embeddings were normalized and used to construct a FAISS vector database. Each stored passage was prefixed with the `passage:` identifier required by the E5 embedding formulation. The resulting vector database stores both the semantic representation of each passage and its associated metadata, including the original document page, required for traceability in the final system output.

2.2 Synthetic patient dataset

To provide a controlled set of clinical scenarios for the development and testing of the proposed system, a synthetic dataset of patient cases was generated with the assistance of GPT-4.5.

The dataset comprises 11 synthetic cases covering three categories: pharyngitis, urinary tract infection (UTI), and an unrelated control case. For both pharyngitis and UTI, five cases were generated, including two cases in which antibiotic treatment is clinically indicated, two cases in which antibiotic treatment is not indicated, and one edge case presenting additional clinical complexity. A further unrelated control case was included to assess the behaviour of the system when the presented condition falls outside the main clinical scenarios considered (Fig. 1).

The dataset was designed to reproduce the type of heterogeneous information that may be available to a general practitioner, so each patient record contains a free-text doctor note describing the presenting complaint and relevant clinical findings, together with structured EHR information. The latter includes demographic characteristics, geographic region, allergies, previous antibiotic prescriptions and comorbidities. Clinical data include reported symptoms, symptom duration, temperature, relevant diagnostic tests and, where applicable, clinical scores such as the Centor/McIsaac score for pharyngitis. Pregnancy status and gestational age were also included in selected cases to represent patient characteristics that may influence antibiotic selection. The patient data structure is shown in Figure 2.

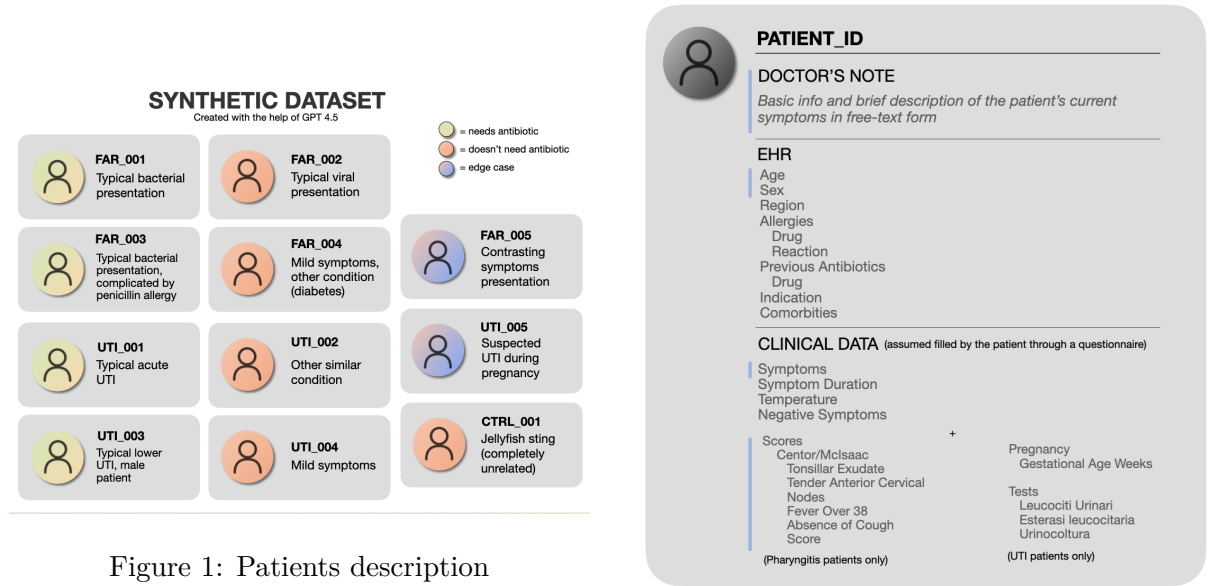


Figure 1: Patients description

Figure 2: Patient data structure

Since the cases are synthetic, the dataset is not intended to represent the prevalence or distribution of real-world diseases. Rather, it provides a controlled environment for developing the system and qualitatively examining its behaviour across representative prescribing scenarios.

2.3 RAG Retrieval

Rather than using the doctor note alone, the query was assembled by combining unstructured and structured patient information, with the free-text doctor note used as the main clinical description. Chosen fields are highlighted in blue in Figure 2. Additional structured information and general treatment-related terms were appended to each query to orient the semantic retrieval towards guideline sections concerning therapeutic management, rather than solely to-

wards descriptions of the disease.

Each query was embedded and used to perform similarity search against the previously constructed vector database. For each case, the 15 highest-ranking chunks were initially retrieved, after which a distance threshold of 0.30 and page-level deduplication were applied. Up to 10 unique guideline pages were consequently retained as evidence for the subsequent reasoning stage.

2.4 Regional Antibiotic Pressure Calculation

To incorporate regional antibiotic consumption into the patient context, AIFA 2024 consumption data (https://www.aifa.gov.it/documents/20142/847578/dati2024_04.12.2025.csv) were used. Records belonging to ATC class J were selected, and antibiotic usage for each region r was calculated as the sum of traceability and reimbursed packages. The resulting value was normalized across regions using min-max normalization:

$$P_r = \frac{U_r - U_{\min}}{U_{\max} - U_{\min}} \quad (1)$$

where U_r is the total antibiotic consumption in region r . The normalized score ranges from 0 to 1 and was subsequently divided into three levels (low, medium, high) using tertiles. The corresponding regional pressure score and level were then associated with each synthetic patient according to the region recorded in the EHR.

2.5 Context Building and Prompt Engineering

The retrieved evidence was combined with the synthetic patient data to construct a structured context for the LLM.

The resulting context was then passed to *Gemini 3.6 Flash*, which was used as the reasoning component of the RAG pipeline. Prompt engineering was used to constrain the model to the information contained in the patient record and the retrieved AIFA recommendations, explicitly discouraging the use of external medical knowledge or unsupported assumptions.

The prompt also explicitly instructed the model to identify the clinical profile, determine the relevant decision criteria in the retrieved evidence, and match these criteria to the patient’s characteristics. Particular emphasis was placed on prioritizing specific recommendations and decision thresholds over generic statements contained in the retrieved text. The prompt also defined separate instructions for pharyngitis and urinary tract infection cases, including the use of the Centor/McIsaac score and relevant UTI clinical features.

Particular attention was given to preventing irrelevant retrieved information from influencing the decision. For example, regional antibiotic-pressure information was not treated as an independent prescribing criterion unless explicitly required by the retrieved guideline evidence. Similarly, the model was instructed to return an undefined decision only when information necessary for applying an applicable guideline rule was genuinely missing.

Finally, the output format was constrained to a fixed JSON schema, containing the antibiotic recommendation, active ingredient (only when explicitly supported by the retrieved evidence), a textual rationale, the guideline pages used for the decision, missing information, and an indicator of evidence sufficiency.

2.6 Pharmaceutical matching

The final stage maps the active ingredient returned by the LLM to corresponding pharmaceutical products using official AIFA data. The system uses the AIFA *Classe A per principio attivo* dataset (https://www.aifa.gov.it/documents/20142/3789005/Classe_A_per_principio_attivo_31-12-2025.csv), which contains active ingredients together with corresponding pharmaceutical denominations and commercial products.

Before matching, both the LLM output and the AIFA active-ingredient fields are normalized by converting text to lowercase, removing Italian diacritics, standardizing whitespace, and normalizing combination separators. The matching procedure first attempts an exact match between the normalized active ingredient and the AIFA field. If no exact match is found, substring-based matching is applied in both directions. Combination therapies are decomposed into their constituent alternatives before matching, while duplicate commercial products are removed.

2.7 Final presentation

The final output is presented through a doctor-facing interface designed to provide a concise overview of the generated clinical decision support. For each case, the interface displays the patient information, clinical note, antibiotic recommendation, selected active ingredient, corresponding pharmaceutical options, reasoning, and the AIFA evidence used to support the decision. An example of the final presentation is provided at the end of the document.

3 Experimental Results

3.1 RAG Retrieval

Overall, the retrieved evidence showed good qualitative relevance across the synthetic cases. Most of the retrieved pages concerned the clinical condition and antibiotic management of the corresponding patient, with the highest-ranked results generally providing the most relevant evidence. Retrieval was slightly less consistent for UTI cases than for pharyngitis cases, as seen in Figure 3. The control case represented the main exception: despite being unrelated to antibiotic prescribing, it returned guideline pages with relatively high similarity scores that were not clinically relevant to the presented condition.

FAR_001	p.50 / 0,233	p.52 / 0,240	p.53 / 0,245	p.49 / 0,253	p.41 / 0,253
FAR_002	p.50 / 0,240	p.53 / 0,247	p.49 / 0,253	p.54 / 0,254	p.51 / 0,260
FAR_003	p.50 / 0,243	p.53 / 0,246	p.51 / 0,262	p.49 / 0,262	p.52 / 0,267
FAR_004	p.53 / 0,246	p.50 / 0,246	p.51 / 0,257	p.49 / 0,267	p.54 / 0,269
FAR_005	p.50 / 0,228	p.53 / 0,240	p.49 / 0,243	p.58 / 0,251	p.54 / 0,253
UTI_001	p.231 / 0,242	p.236 / 0,253	p.233 / 0,253	p.230 / 0,256	p.232 / 0,260
UTI_002	p.233 / 0,270	p.231 / 0,275	p.230 / 0,278	p.177 / 0,281	p.109 / 0,282
UTI_003	p.231 / 0,242	p.233 / 0,252	p.236 / 0,259	p.230 / 0,264	p.232 / 0,270
UTI_004	p.233 / 0,265	p.177 / 0,277	p.231 / 0,278	p.232 / 0,279	p.64 / 0,280
UTI_005	p.231 / 0,247	p.233 / 0,249	p.230 / 0,263	p.236 / 0,263	p.109 / 0,269
CTRL_001	p.61 / 0,266	p.165 / 0,272	p.63 / 0,273	p.185 / 0,277	p.64 / 0,278

Figure 3: Top 5 results by L2 distance, strikethrough indicates irrelevant page

3.2 Clinical Decision Results and Qualitative Evaluation.

The clinical decision-making performance of the proposed system was qualitatively evaluated by three medical experts with complementary professional backgrounds: a general practitioner, a healthcare manager, and an infectious disease specialist (infettivologist). Each expert independently assessed the model-generated responses on a scale from 1 to 10, considering their clinical appropriateness, completeness, consistency, and relevance to the presented patient scenario. The mean score assigned to the model responses was 8.18/10, indicating an overall positive assessment of the system’s clinical reasoning capabilities. The individual evaluations, including the assigned scores and the experts’ qualitative reasoning, are reported in Figure 4.

= IMPRECISION = CORRECT = INCORRECT OR OMITTED				General Practitioner Healthcare Manager Infettivologist		
FAR_001	Amoxicillina	Info on further tests	Posology and treatment duration	8	9	6
FAR_002	No Antibiotic	Age in Centor	Directives for doctor	8	10	9
FAR_003	Antibiotic	Claritromicina Cefalexatina	Patient age Swab needed	8	9	5
FAR_004	No Antibiotic	Patient age	MMG mention	7	10	9
FAR_005	Not enough info	Swab test prescription	RF mention	5	7	4
UTI_001	Nitrofurantoina	Amoxicillina+acido clavulanico	Additional antibiotics (Fosfomicina)	8	10	6
UTI_002	No Antibiotic			8	9	10
UTI_003	Antibiotic	Nitrofurantoina	Antibiogram Lifestyle	8	10	5
UTI_004	No Antibiotic			8	10	10
UTI_005	Amoxicillina+acido clavulanico	Pregnancy test considerations		8	10	6
CTRL_001	No Antibiotic			10	10	10

Figure 4: Expert evaluation of the model’s clinical decision-making. The figure reports the scores assigned by the three medical experts together with their qualitative reasoning.

Overall, the model demonstrated strong performance in the fundamental distinction between cases requiring antibiotic treatment and cases in which antibiotics were not indicated. In particular, it correctly identified the control case as clinically irrelevant to the antibiotic decision-making task. It also appropriately refrained from providing a treatment recommendation for the pharyngitis edge case, recognizing that the available clinical information was insufficient to support a reliable decision. These results suggest that the retrieval and reasoning components were generally effective in supporting the high-level clinical decision of whether antibiotic treatment was warranted.

However, the evaluation revealed a consistent difference between cases requiring and not requiring antibiotic treatment. The model generally performed better on cases in which antibiotics were not indicated, whereas its responses were less consistent when a specific antibiotic treatment had to be recommended. In these cases, several clinically relevant aspects were either overlooked or incompletely addressed. For example, in `uti_003`, the model did not adequately account for the clinical implications associated with the male patient presentation. In other cases, the recommendations did not consistently specify appropriate posology or treatment du-

ration. The model also omitted some recommended swab tests in **far_003** and **far_005**, while in **uti_001** it failed to include additional relevant treatment options, such as fosfomycin. Furthermore, some of the generated antibiotic lists were considered excessively long, suggesting that the model may prioritize completeness over clinical selectivity when multiple possible treatment options are available.

A particularly relevant limitation was observed in the UTI edge case, where the model generated an antibiotic prescription without adequately considering pregnancy-related safety requirements. This finding highlights the importance of explicitly incorporating patient-specific contraindications and mandatory diagnostic or screening information into the reasoning process.

The expert evaluation also highlighted limitations in the realism of some synthetic patient profiles, particularly with respect to the age of patients in the pharyngitis cases and the absence of pregnancy-related information or mandatory pregnancy testing considerations in the UTI cases. These limitations are relevant because incomplete or unrealistic patient profiles can directly affect the quality of the model’s clinical reasoning. The overall realism of the synthetic patient cases received a mean score of 8/10 from the medical experts, as shown in Figure 5.

In addition to the case-level assessment of model responses, the experts provided an overall evaluation of the proposed system, assigning a mean score of 9.33/10, as reported in Figure 6. This high overall score indicates that, despite the limitations identified, the system was perceived as a strong overall solution for the considered clinical decision-support task.

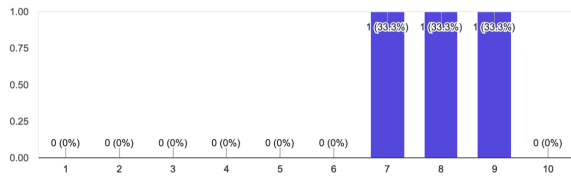


Figure 5: Expert evaluation of the realism of the synthetic patient cases.

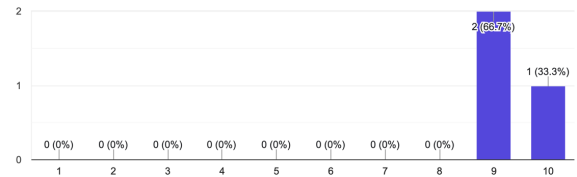


Figure 6: Expert evaluation of the overall model.

4 Concluding Remarks

The qualitative evaluation indicates that the proposed system provides a strong baseline for antibiotic decision support, particularly in distinguishing cases in which antibiotic treatment is not required from those in which it may be appropriate. The high evaluation score obtained by the model suggests that its overall clinical reasoning is promising; however, the analysis also demonstrates that accurate antibiotic prescribing requires more than identifying the correct treatment category.

In particular, future improvements should focus on strengthening patient-specific reasoning, ensuring that relevant demographic and safety information is explicitly considered, and improving the completeness of treatment recommendations, including posology, treatment duration, diagnostic testing, and alternative therapeutic options. The control case also highlighted limitations in retrieval, as highly similar pages could be retrieved despite being unrelated to the clinical scenario. These observations suggest that retrieval quality and explicit verification of patient-specific characteristics remain important challenges.

Overall, the results support the potential of the proposed RAG-based approach while demonstrating the need for further refinement and more extensive expert-based evaluation before considering its use in real-world clinical settings.

UTI_001

Paziente

Età: 26 Sesso: femmina Regione: Toscana

Allergie: nessuna Gravidanza: no

Sintomi: disuria acuta, aumento della frequenza della minzione, urgenza minzionale, dolore o disagio nella parte inferiore dell'addome

Temperatura: 36.7 °C Durata: 2 giorni

Test: leucociti_urinari: True; esterasi_leucocitaria: True; urinocultura: None

Nota clinica

Paziente di 26 anni presenta da 2 giorni disuria acuta, aumento della frequenza della minzione e urgenza minzionale. Riferisce lieve dolore e disagio nella parte inferiore dell'addome. Non riferisce altri disturbi genito-urinari. Il quadro viene valutato nel contesto di una sospetta infezione delle vie urinarie inferiori.

Valutazione

Decisione: antibiotico raccomandato

Principio attivo / opzioni: Nitrofurantoina (o Amoxicillina+acido clavulanico)

Ragionamento

La paziente presenta una manifestazione clinica compatibile con infezione delle vie urinarie inferiori (disuria acuta, aumento della frequenza e dell'urgenza minzionale, dolore addominale inferiore da 2 giorni) e riscontri di laboratorio positivi (leucociti urinari ed esterasi leucocitaria positivi). Secondo le linee guida recuperate, il trattamento antibiotico è raccomandato in presenza di manifestazione clinica compatibile associata a test positivo per leucociti urinari/esterasi leucocitaria.

Informazioni mancanti: nessuna

Farmaci corrispondenti

Nitrofurantoina: NEOFURADANTIN, NITROFURANTOINA

Amoxicillina/acido clavulanico: ABBA, ABIACLAV, ACADIMOX, AKLAV, AMOCLA, AMOXICILLINA E ACIDO CLAVULANICO P-CARE, AMOXICILLINA E ACIDO CLAVULANICO, ANIVAL, AUGMENTIN, AVEGGIO, CLAVOMED, CLAVULIN, HOMER, KLAVUX, KRUXADE, MONDEX, NEODUPLAMOX, POSMOX, PURICLAV, STEMOX, HOMERDELIX

Evidenza AIFA utilizzata

Pagine selezionate: 230, 231, 232

Pagina 231: Infezione delle vie urinarie inferiori. Manifestazione clinica: disuria acuta (<1 settimana), aumento dell'urgenza e della frequenza della minzione, dolore o disagio addominale inferiore e talvolta ematuria marcata. Nelle donne deve essere esclusa in primo luogo una fonte vaginale dei sintomi.

Pagina 230: Il capitolo affronta la cistite acuta acquisita in comunità. Nitrofurantoina orale per 5 giorni è il principale trattamento raccomandato per le infezioni del tratto urinario inferiore. L'urinocultura deve essere considerata nei soggetti ad alto rischio di infezioni complicate o in caso di infezioni ricorrenti.

Pagina 232: Trattamento antibiotico raccomandato in caso di manifestazione clinica compatibile e test positivo (leucociti urinari/esterasi leucocitaria positivi o urinocultura positiva). Gli antibiotici includono Amoxicillina+acido clavulanico e Nitrofurantoina.

Figure 7: Example of the final clinical decision-support output for case UTI_001.